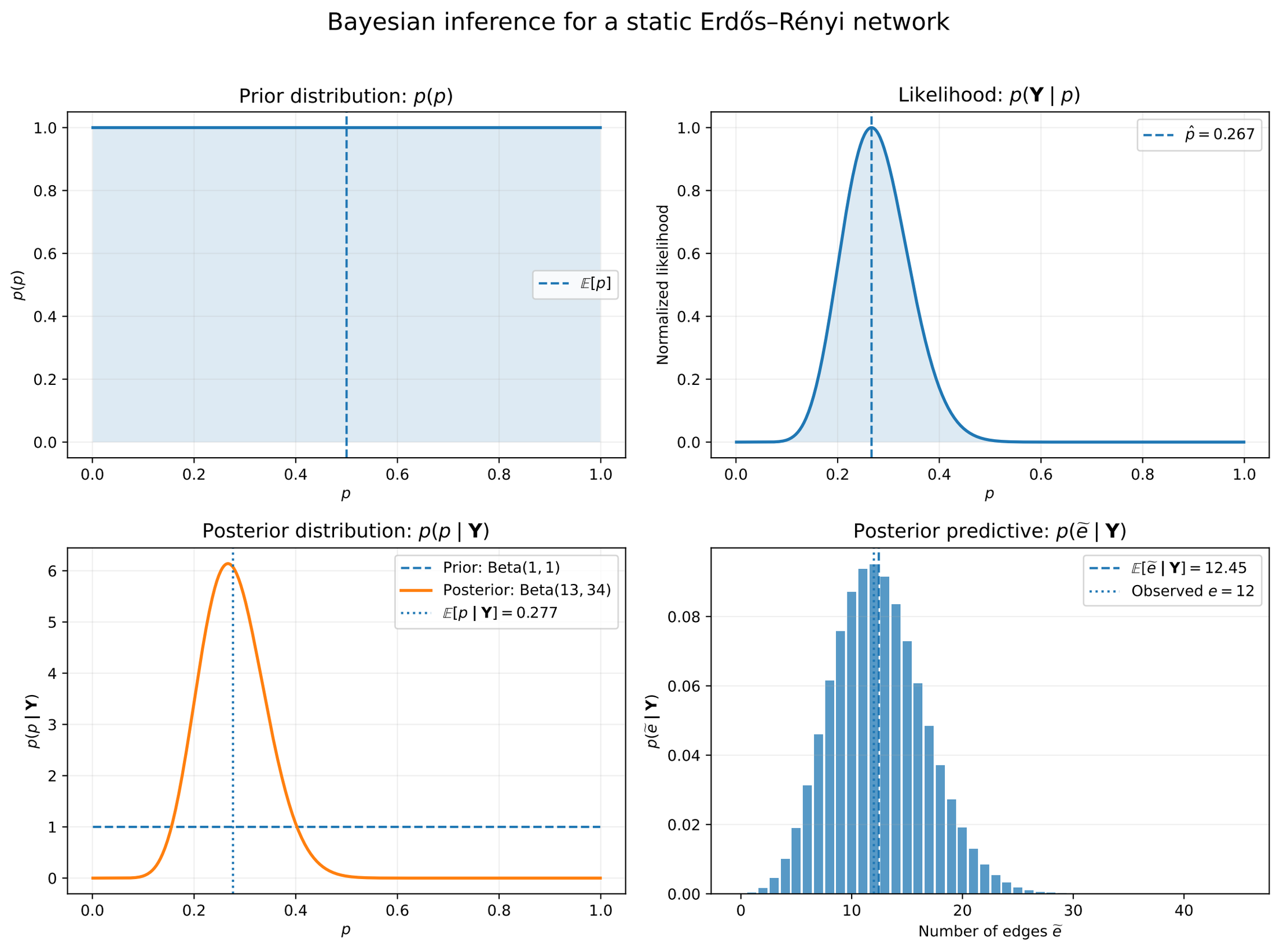

Bayesian inference for a static Erdős–Rényi network
Prior distribution: $p(p)$
$p(p)$
p
$\mathbb{E}[p]$
Likelihood: $p(\mathbf{Y} \mid p)$
Normalized likelihood
p
$\hat{p} = 0.267$
Posterior distribution: $p(p \mid \mathbf{Y})$
$p(p \mid \mathbf{Y})$
p
Prior: Beta(1, 1)
Posterior: Beta(13, 34)
$\mathbb{E}[p \mid \mathbf{Y}] = 0.277$
Posterior predictive: $p(\tilde{e} \mid \mathbf{Y})$
$p(\tilde{e} \mid \mathbf{Y})$
Number of edges $\tilde{e}$
$\mathbb{E}[\tilde{e} \mid \mathbf{Y}] = 12.45$
Observed $e = 12$